

Owen Zhu

📍 Toronto, ON

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EDUCATION

University of Waterloo | Bachelor of Computer Science, Honours Co-op

Waterloo, ON | Sep 2020 – Apr 2025

GPA: 3.7/4.0 | Graduating with Distinction

EXPERIENCE

Amazon Robotics 📁 | Software Development Engineer

Toronto, ON | July 2025 – Present

- Acted as **Technical Lead** for global initiatives, owning the full SDLC from development to **Change Management** execution during code freezes; received **VP-level recognition** for flawless global rollouts.
- Spearheaded the strategic deprecation of a legacy data model service, dismantling obsolete pipelines and NLBs while collaborating with client teams to migrate critical **S3/DynamoDB** data with zero downtime.
- Drove **Operational Excellence** during bi-monthly on-call rotations, leading the resolution of **critical production incidents** and minimizing operational toil to maintain 99.99% availability for fleet services.
- Designed event-driven microservices for robotic fleet management using **Kotlin** and **AWS Lambda**, processing high-throughput telemetry via SQS/SNS with **idempotent consumers**.

Faire 📁 | Android Engineer Intern

Toronto, ON | Sep 2024 – Dec 2024

- Led the migration to type-safe **Compose Navigation**, designing a custom routing system to pass complex arguments with optimized serialization, reducing navigation latency by **200ms**.
- Authored a comprehensive Engineering Design Document to standardize navigation patterns, researching and implementing type-safe arguments to ensure scalability and developer velocity.
- Engineered the Android frontend for **experimental AI features**, transforming design concepts into functional implementations to introduce multimodal capabilities for the retailer app.
- Enhanced the custom MVVM architecture using **RxJava** to streamline reactive data flows, significantly improving code maintainability for complex retailer workflows.

Intuit 📁 | Software Engineer Intern

Toronto, ON | Sep 2023 – Dec 2023, May 2024 – Aug 2024

- Built the 'Document Checklist' widget in **Native Android**, replacing a legacy WebView to reduce load times from **6.0s to 1.5s**, streamlining the document upload flow for **3M+ monthly users**.
- Designed and deployed exclusive features (swipe-to-delete, bulk uploads, cloud integration) that increased user conversion rates, validated through production **A/B testing**.
- Reduced manual testing time by implementing automated UI tests using **Appium**, leveraging XCUITest (iOS) and UIAutomator2 (Android) to ensure cross-platform stability.

theScore 📁 | Software Developer Intern

Toronto, ON | Jan 2023 – Apr 2023

- Delivered enhancements to 10+ screens including the new 'Teasers' bet type, leveraging Kotlin and **Apollo GraphQL** to drive user engagement.
- Refactored legacy code to remove **5000+ unused lines**, reducing module compile times by **10%** and improving developer velocity.

Innovapost 📁 | Android Developer Intern

Ottawa, ON | Apr 2022 – Aug 2022

- Developed a proof-of-concept NFC-based SSO login system, reducing authentication time by **70%** and streamlining access for thousands of logistics employees.
- Strengthened security by implementing **AES-256** encryption for credential storage and integrating Android's Keystore system.

PROJECTS

Java Compiler (C++) 📁

- Built a compiler for Java 1.3 from scratch using **C++**, employing flex/bison for lexing and parsing, and drawing inspiration from Clang for AST design and construction.
- Implemented advanced compiler optimizations including **constant folding** and **dead code elimination** to significantly enhance executable performance.

SKILLS

Languages: Kotlin, Java, Python, TypeScript, SQL, C/C++, Go, Rust, Swift

AWS / Cloud: Lambda, DynamoDB, AppConfig, ECR, CDK (IaC), CloudFormation, SQS/SNS, IAM, CloudWatch

Practices & Ops: System Design, Microservices, Operational Excellence, Distributed Systems, CI/CD, Agile

Frameworks & Tools: Spring Boot, React, Node.js, Jetpack Compose, Apollo GraphQL, Docker, Git, Postman